

Yee vs FCI

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1 Introduction

In this report, electromagnetic (EM) problems are addressed using two distinct numerical solvers: Yee and FCI FDTD. These schemes are compared in numerical accuracy and computational efficiency. Section 1.1 gives a short overview of how the codes can be used to perform simulations. Section 2.1 validates these codes by comparing them to the analytical frequency response in free space. Then, in section 2.2, experiments with a Fabry-Perot etalon are performed to further examine the difference between Yee and FCI. Lastly, in section 2.3, frequency analysis of a drude material is performed. All experiments were conducted using a Gaussian-modulated sinusoidal source:

$$J(t) = J_0 \sin(\omega_c t) \exp\left(-\frac{t-t_c}{2\sigma^2}\right) \quad (1)$$

This source excites the frequency band $\omega \in [\omega_c - \frac{3}{\sigma}, \omega_c + \frac{3}{\sigma}]$ and by choosing $\sigma = \frac{3}{\omega_c}$ this frequency band reduces to $[0, 2\omega_c]$.

1.1 Code description

To solve problems using the Yee-FDTD scheme, the python code `Yee_class.py` was written. This code contains the python class Yee, designed to solve and analyse problems using the Yee-FDTD scheme. The most important functions of this class are listed in table 1. A similar file were created for analysis using the FCI scheme (`FCI_class.py`).

Name	Arguments	Short description
add_source	(xs,ys,J0,tc,width,Wc)	Adds a current source at indices i=xs and j=ys with amplitude J0, peak time tc, width w and angular frequency Wc
add_recorder	(xr,yr)	Adds a recorder at indices i=xr and j=yr
add_material	(x_s,x_e,y_s,y_e,eps_r, mu_r, sigma)	Adds rectangular material with vertices at indices (x_s,y_s) and (x_e,y_e) with $\epsilon_r = \text{eps}_r$, $\mu_r = \text{mu}_r$ and $\sigma = \text{sigma}$.
add_drude_material	(x_s,x_e,y_s,y_e,eps_r, sigma_DC, gamma)	Adds rectangular drude material with vertices at indices (x_s,y_s) with $\epsilon_r = \text{eps}_r$, $\sigma_{DC} = \text{sigma_DC}$ and $\gamma = \text{gamma}$.
update_loop	None	Solves problem at all timesteps Nt
show_recorder	None	Plots E_z at recorder position for all timesteps Nt
animate	None	Animates E_z on entire grid for all timesteps Nt
analytical_solution	(p_all=True,f_lim=None)	Returns ω , $H_{ana}(\omega)$, $H_{num}(\omega)$ (see section 2.1) for frequencies $\omega \in [0, f_lim]$

Table 1: Most important functions of classes Yee and FCI

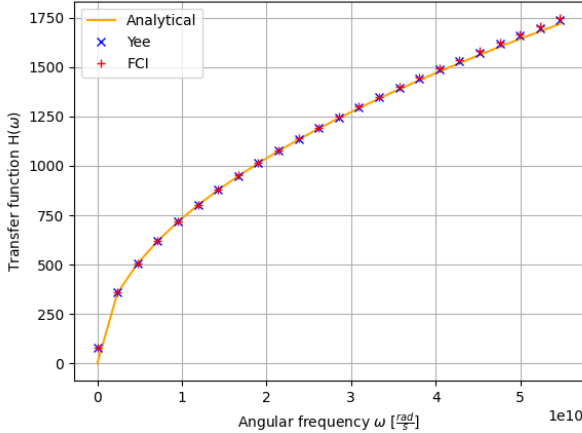
2 Results and Discussions

2.1 Comparison with an analytic solution

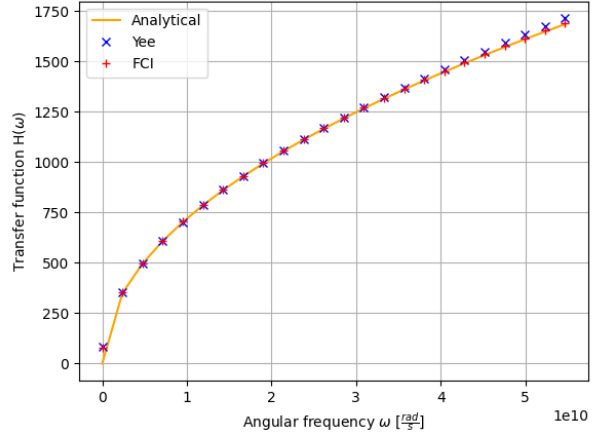
First, the ability of both the Yee and FCI schemes to reproduce the analytic frequency response $H(\omega)$ of a 2D-TM electromagnetic wave will be evaluated. This frequency response $H(\omega)$ is given by:

$$H(\omega) = -\frac{\omega\mu}{4}H_0^{(2)}\left(k_0\sqrt{(x_r - x_s)^2 + (y_r - y_s)^2}\right) \quad (2)$$

where (x_s, y_s) and (x_r, y_r) are the coordinates of the source and the recorder, respectively. In the python file `Comparison_analy.py`, the comparison between Yee and FCI is performed. A sinusoidal source with angular frequency $\omega_c = 2.8 \cdot 10^{10}$ rad/s is used, and consequently, only the frequency range $\omega \in [0, 2\omega_c]$ is plotted. The results of both the x-direction and diagonal propagation are shown in figures 1a and 1b, respectively.



(a) Horizontal (x-direction)



(b) Diagonal

Figure 1: Comparison analytical and numerical frequency response for Yee and FCI in the vertical and diagonal direction

Both schemes follow the analytical solution well in the frequency interval of the source, which serves as the first validation for both schemes. It can be seen that at high frequencies, the deviation of the analytical solution is higher for propagation in the x-direction than for diagonal propagation. This is expected for both schemes, since the dispersion is expected to be the highest in the coordinate directions. It can also be seen that for higher frequencies, FCI is more accurate than Yee in the diagonal direction, again confirming the theory. Another advantage of FCI is its unconditionally stable behavior. However, this comes at a cost. FCI is much more time consuming than Yee due to it being an implicit scheme. Figure 2 shows the comparison between Yee and FCI for an experiment with $N_x = N_y = 401$ and $N_t = 4000$.

2.2 Fabry-Perot

In this section, the results of reflection experiments on a Fabry-Perot etalon (thin dielectric) are discussed. These experiments were performed in the python file `FP_test.py`. The normalized power reflection coefficient $R(\omega)$ is measured for slabs of different thicknesses d . In these experiments, the source and recorder were placed on top of each other. Thus, to obtain the reflected signal, the source signal can simply be subtracted from the recorder.

$$R(\omega) = \left| \frac{E_{z,refl}(\omega)}{E_{z,appl}(\omega)} \right|^2 = \left| \frac{E_{z,rec}(\omega) - E_{z,appl}(\omega)}{E_{z,appl}(\omega)} \right|^2 \quad (3)$$

Here, $E_{z,rec}$ and $E_{z,appl}$ are the Fourier transforms of the recorded and applied electric fields. The results of these experiments are given in figure 3.

Due to interference phenomena, it is known that maxima will be seen for the frequencies where $d = m\frac{\lambda}{2}$, with m an integer. This leads to a peak spacing Δf :

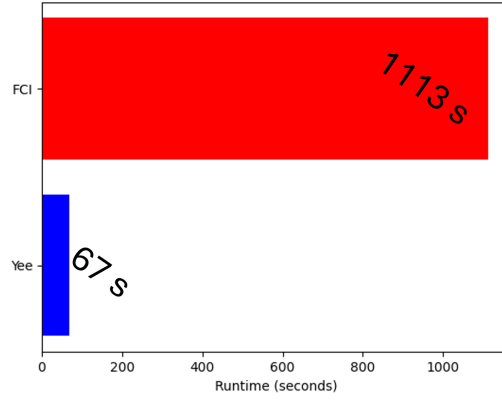


Figure 2: Time comparison between Yee and FCI for $N_x = N_y = 401$ and $N_t = 4000$

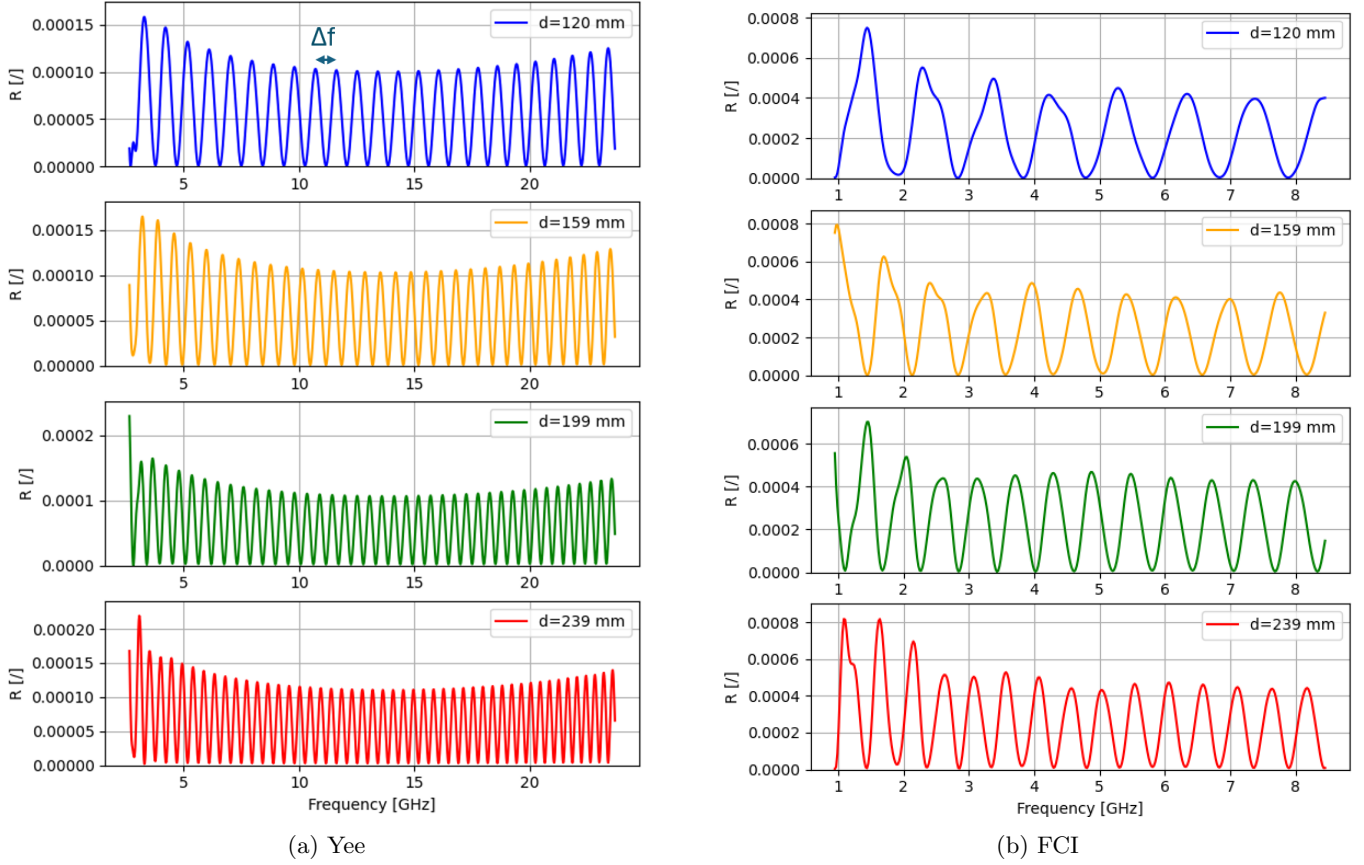


Figure 3: Normalized power reflection coefficient in Yee and FCI for different thicknesses d

$$\Delta f = \frac{c}{2nd} \quad (4)$$

This analytical peak spacing is indeed found in the simulations. Figure 4 compares the analytical interference peaks with the numerical reflection coefficient for both the Yee and FCI schemes. At low frequencies, the peak positions correspond to the analytical predictions, but for higher frequencies, dispersion effects become clear. In the Yee scheme, the numerical waves propagate slower than the real waves, resulting in reduced peak spacing. The opposite is true for the FCI scheme.

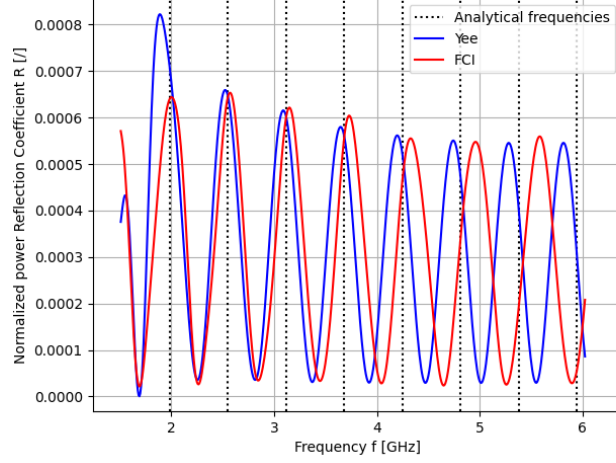


Figure 4: Comparison between numerical and analytical frequency peaks for $d = 200\text{mm}$

2.3 Drude material

In this section, the transmission through a Drude material was investigated and compared to a conventional metal. These experiments were performed in the python file `drude.py`. A Drude material represents more complex metals, for which the frequency-dependency of the conductivity σ is assumed to behave as follows:

$$\sigma(\omega) = \frac{\sigma_{DC}}{1 + j\omega\gamma} \quad (5)$$

A Drude material has conductivity σ_{DC} at low frequencies and becomes transparent at high frequencies. In the simulation, the DC-conductivity is chosen such that, for a conventional metal, only a part of the applied wave is reflected. Furthermore, in the FCI simulation, the grid is refined in the material region in order to obtain more accurate results. The parameter γ is chosen to be $1/\omega_c$, such that the material becomes transparent within the applied frequency band, since the cut-off frequency is given by $\omega_s = \frac{1}{\gamma} = \omega_c$. This is the frequency where the amplitude of the conductivity $|\sigma(\omega_s)| = \frac{\sigma_{DC}}{\sqrt{2}}$. In figure 5, the simulation indeed shows this behavior.

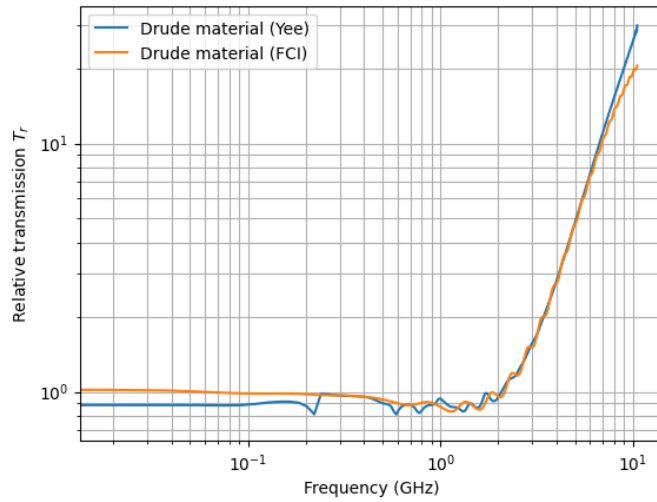


Figure 5: Transmission of a Drude material relative to a normal conductor

The relative transmission coefficient T_r , given by formula 6, is plotted in figure 5.

$$T_r = \left| \frac{E_{t,drude}}{E_{t,simple}} \right|^2 \quad (6)$$

The linear behavior of the transmission coefficient T_r , expected for a Drude material, is clearly present. This indicates that these materials are modelled correctly in both the Yee and FCI schemes. For this experiment, a grid refinement was performed in FCI at the material, to improve accuracy and maintain computational efficiency.

3 Conclusion

In this project, both Yee- and FCI-FDTD-schemes were successfully implemented to simulate electromagnetic waves. Both methods match the analytical solution in the applied frequency range. At high frequencies, a small deviation from the analytical solution is observed, which is smallest in the diagonal direction, as predicted by the numerical dispersion relation of both schemes. Confirming the theory, these deviations at high frequencies are smaller in the FCI scheme compared to the Yee scheme. It can be concluded that the FCI method is more accurate, but this comes at a high computational cost.

Both codes were used to simulate a Fabry-Perot etalon. FCI and Yee schemes were both able to reproduce the expected peaks in the power reflection coefficient. However, due to dispersion, the peaks were slightly shifted to the left in the Yee scheme and to the right in the FCI scheme. This is again due to numerical dispersion, giving a positive phase error for FCI and a negative phase shift for Yee.

Lastly, a Drude material was effectively simulated and showed the expected frequency behavior. Grid refinement in FCI was effectively used in this simulation, showing that more efficient computation is possible without losing too much accuracy.